11 **(a)** Define the half-life of a radioactive source.

.....

..... [1]

(b) A protactinium (Pa) nucleus decays into a uranium (U) nucleus by the emission of a beta particle (β-particle).

(i) Complete the nuclear equation for the decay.

$$^{234}_{91}\text{Pa} \rightarrow$$

[3]

(ii) State the change that occurs in the nucleus during the decay.

..... [1]

(c) A different element decays by the emission of an alpha particle (α-particle).

Give **two** reasons why α-particles are more strongly ionising than β-particles.

1 .....

.....

2 .....

.....

[2]

[Total: 7]

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